

Supplementary Material: Text-Driven Inverse Tone Mapping for Compressed SDR Videos via Multi-Modal Description

A Analysis

1.1 Importance of text-driven

There are three main reasons why we adopt textual representations as an intermediate modality. First, our model focuses on ITM for compressed SDR videos. Therefore, it is necessary to obtain compression-related information from SDR videos. Many existing approaches (1), (2) extract such information from video metadata. Following this line of work, we also retrieve compression information from metadata. However, directly encoding metadata into discriminative features is nontrivial. We also attempted to use an MLLM to identify compression information directly from video content, but found that this approach was unreliable. We present an example of MLLM-generated compression information in Fig. 1. As shown in Fig. 1, the MLLM produces incorrect results, including an erroneous compression format (H.264 $\rightarrow$ H.265) and an inaccurate QP estimate (42 $\rightarrow$ 22–28). Consequently, we adopt a simple yet effective strategy: we extract compression information from metadata using software tools (or manual inspection), convert it into textual form, and then encode it into feature representations using a text encoder. Since our task is ITM, compression information alone is insufficient. We also require task-relevant cues, including scene semantics and low-level attributes, to better guide the ITM network. To ensure that all these heterogeneous sources of information reside in a unified embedding space, we convert them into textual form and encode them through the same text encoder.

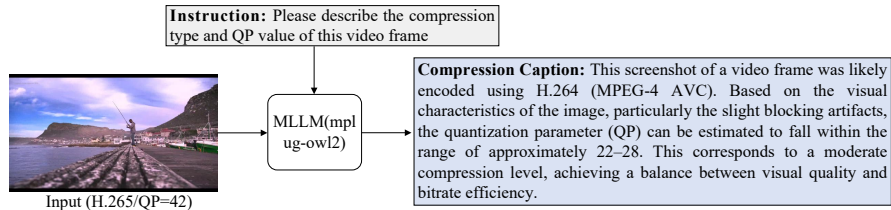


Figure 1: An example of MLLM-generated compression information.

Second, by converting visual and technical information into a textual format, we can intuitively monitor whether the generated descriptions are accurate. This transparency allows us to identify and filter out "hallucinations" or incorrect information, preventing the model from being guided by erroneous inputs—a safeguard that is not possible with "black-box" visual features.

Third, in our future work, we aim to extend the current model into an interactive system. Using a textual intermediate allows for human intervention, where users can manually modify or refine the descriptions to enhance the controllability and personalization of the ITM process.

1.2 Robustness of text-driven

First, an important reason for adopting the text-driven pipeline is that the compression information extracted from metadata is already in textual form. To enable multimodal feature alignment across all information sources, we likewise convert semantic information and low-level attribute information into text and encode them into a shared feature space using the same text encoder.

Second, as shown in prior work (3), the sliding manner for MLLM-based caption generation produces high-quality outputs, which indicates that our generated captions have a relatively low probability of containing hallucinations or inaccurate descriptions.

Third, we manually examined the descriptions generated for our test subset to directly assess whether the descriptions were affected by hallucinations. Specifically, the test subset consists of 233 videos, each containing 10 consecutive frames, resulting in a total of 2,330x4 descriptions. For efficiency, we randomly selected the descriptions corresponding to one frame from each video, yielding 233x4 descriptions for manual inspection. Our observations indicate that the generated descriptions are reliable and exhibit minimal hallucination. However, we cannot entirely rule out the possibility that hallucinations may exist in the descriptions of the frames that were not sampled. In future work, we will design validation mechanisms based on mathematical models or lightweight end-to-end networks to identify or filter out hallucinated or incorrect descriptions.

Finally, considering that textual descriptions are easier to validate or examine than visual features, if the quality of the reconstructed iTM videos does not meet expectations, users can first examine whether the generated textual descriptions contain hallucinations or inaccuracies. Once such issues are identified, users can either manually correct the descriptions or leverage a MLLM to regenerate accurate ones, thereby enabling the reconstruction of higher-quality iTM videos. This also provides a promising direction for our future work: developing a human-in-the-loop interactive iTM video reconstruction model.

Overall, although inaccurate textual descriptions may degrade the performance of our model, such descriptions occur with relatively low probability. Therefore, our text-driven pipeline remains both effective and robust.

2. Semantic understanding of MLLM for low-level attributes

2.1 Grounding of the semantic understanding

This can be attributed to three main reasons: first, these low-level attributes

are not unique to the ITM task. They are also widely involved in other computer vision tasks, such as image/video restoration, image/video generation, and image/video quality assessment. Therefore, after being trained on these related tasks, MLLMs naturally acquire the ability to describe such low-level attributes. Second, we utilize an MLLM fine-tuned on the Q-Instruct (4) dataset. Q-Instruct dataset is explicitly curated for low-level visual perception and contains a vast array of samples with detailed annotations of image quality and diverse low-level degradations. This domain-specific training enables the model to bridge the gap between raw visual signals and precise physical descriptors with high reliability. Third, we employ meticulously designed instructions that constrain the MLLM to output specific descriptions, ensuring that the generated priors are consistent with the actual degradation state of the SDR input rather than being arbitrary semantic guesses. With these three reasons, we believe that the MLLM can reliably comprehend the semantics of “overexposed areas” and “texture details”. We provide two representative examples in Fig. 2 to

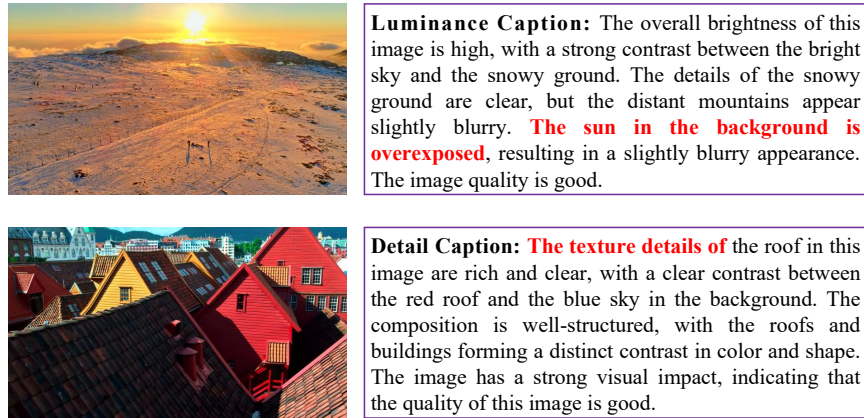


Figure 2: Two examples of luminance and detail captions.

illustrate this capability. As illustrated in Fig. 2, the generated captions explicitly contain semantic information such as “overexposed areas” and “texture details”, demonstrating that the fine-tuned MLLM can effectively characterize low-level attributes.

2.2 Reliability of the semantic understanding

First, to ensure that compression information, semantic information, and low-level attributes reside within a unified embedding space, we represent all such information in a textual format. By leveraging the specialized capabilities developed through the aforementioned fine-tuning, the MLLM acquires a nuanced understanding of low-level attributes. Consequently, with specifically designed instructions, the MLLM can then generate explicit descriptions of these low-level attributes. Second, By transforming low-level attributes into textual

descriptions, we are able to directly and intuitively evaluate the accuracy of the generated outputs. This transparency allows us to identify and filter out incorrect information, preventing the model from being guided by erroneous inputs—a safeguard that is not possible with "black-box" visual features. Third, in our future work, we aim to extend the current model into an interactive system. Using a textual intermediate allows for human intervention, where users can manually modify or refine the descriptions to enhance the controllability and personalization of the ITM process. Finally, through large-scale multi-task training, the MLLM has acquired a generalized understanding of physical concepts such as "overexposure" and "texture," enabling it to adapt to SDR content across diverse scenes and distortion types without task-specific redesign. For these reasons, we leverage the MLLM’s semantic understanding as a reliable prior.

3. Importance of SDR semantic information for HDR videos

There are three main reasons for why SDR semantic information are effective for HDR videos. First, HDR video reconstruction is inherently an ill-posed problem, especially in severely over-/under-exposed regions where detail information is lost. In such cases, purely low-level cues are insufficient, and high-level semantic priors are necessary to guide plausible reconstruction. For example, knowing whether a region corresponds to sky, human skin, or artificial light provides strong cues for reasonable luminance and color recovery. Second, although SDR and HDR differ in dynamic range, the underlying semantic structure and scene layout remain largely preserved in the LDR domain. Object boundaries, spatial arrangements, and material categories are still observable in SDR inputs, enabling reliable extraction of semantic information that can serve as effective priors for HDR video reconstruction. Third, in our model, semantic information is not directly used to HDR video reconstruction. Instead, it is used to guide the ITM network by dynamic parameter estimation and feature interactions. This design allows semantic cues to guide feature adaptation while preserving the dominance of pixel-level reconstruction, thus avoiding over-reliance on high-level descriptions.

B Discuss of low-level attributes

1. Grounding of the choice of three low-level attributes

The selection of luminance, color, and texture details as the three core attributes is fundamentally driven by the primary technical requirements of the ITM task, which can be decomposed into the following three parts: 1) Luminance Mapping: The core of ITM is the expansion of dynamic range from SDR (typically 0-100 nits) to HDR (0-1000+ nits). The luminance component serves as the decisive factor in this mapping process, guiding the intensity scaling and peak brightness restoration. 2) Gamut Mapping: Transitioning from the BT.709 to the BT.2020 color space is essential to represent a wider range of colors. The color attribute provides critical priors that ensure vivid and accurate chromatic

expansion without introducing unnatural shifts. 3) Detail Compensation: Due to limited dynamic range, SDR content often suffers from “clipping” in highlight and shadow regions, leading to a loss of information. Texture details play an indispensable role in guiding the local compensation module to reconstruct these lost details and suppress potential artifacts. Therefore, these three low-level attributes are widely adopted in ITM task. This viewpoint is also validated in (5) and (6).

2. More examples of captions for three low-level attributes

In this section, the detailed procedures of three low-level attributes are shown in Fig. 3, 4, 5. In addition, we present more examples of captions for three low-level aspects in Fig. 6. These operations are also available in our code. We have released the full code, pre-trained models, and database on GitHub.

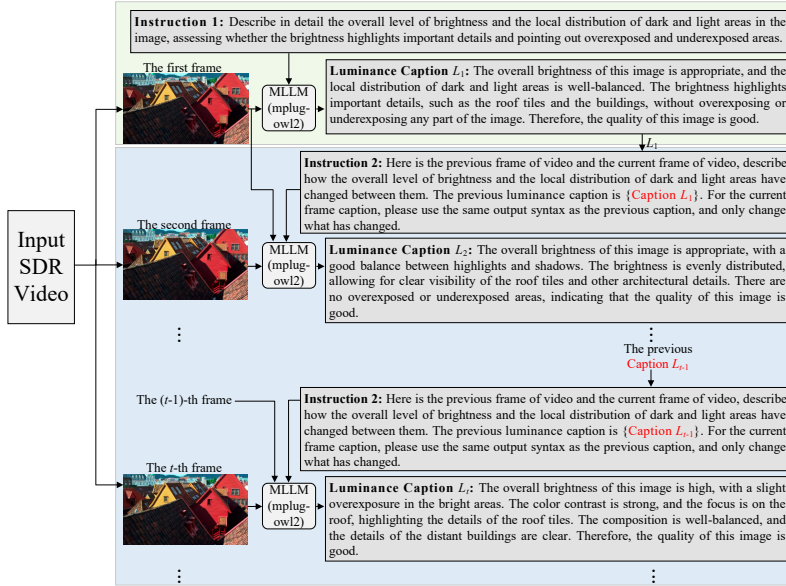


Figure 3: Flowchart for obtaining luminance caption.

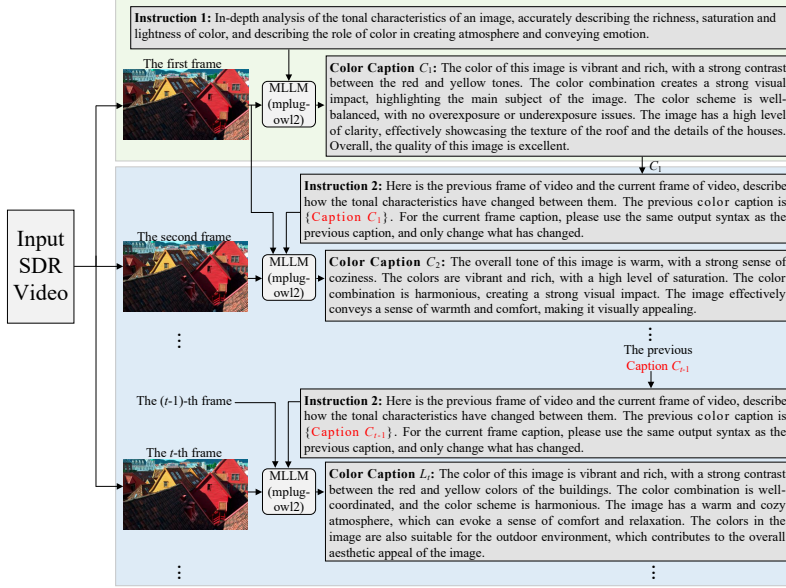


Figure 4: Flowchart for obtaining color caption.

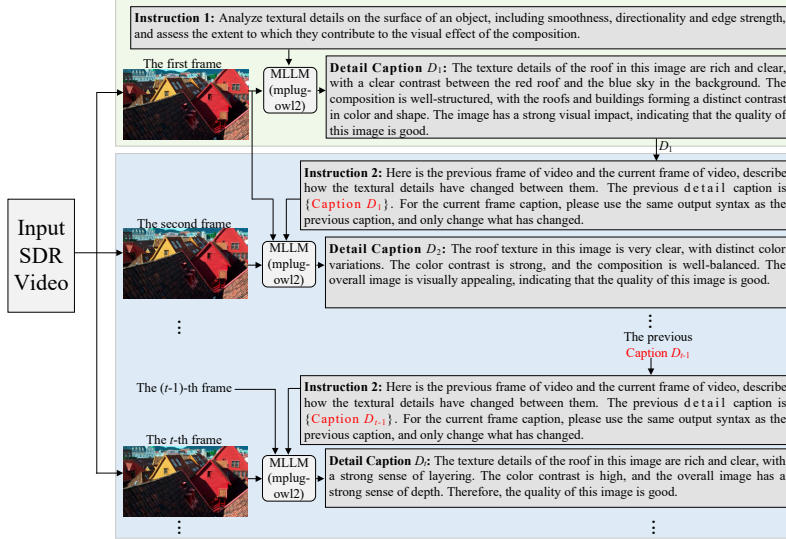


Figure 5: Flowchart for obtaining detail caption.

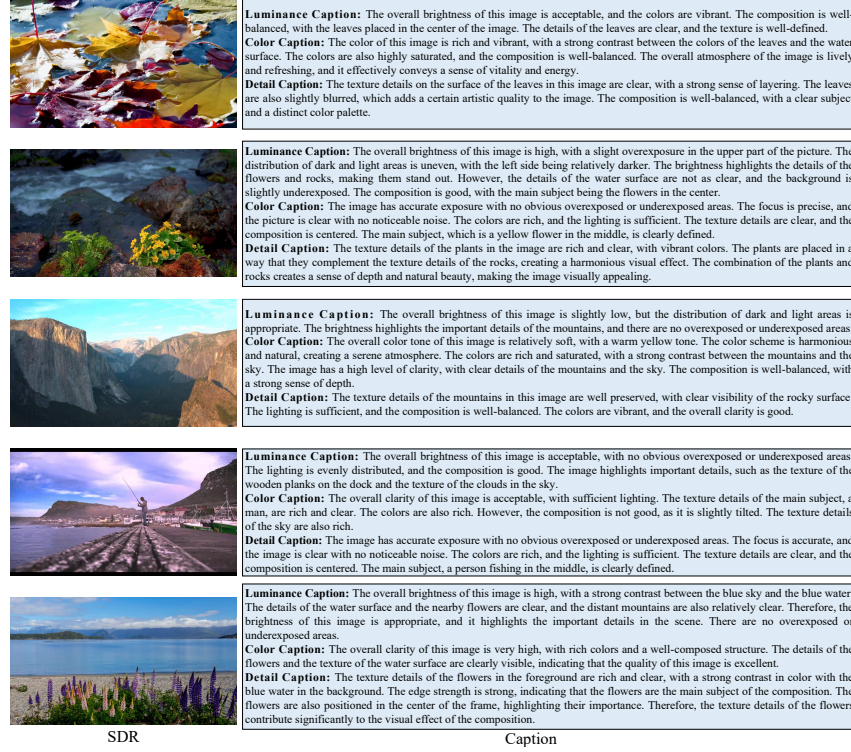


Figure 6: Captions of three low-level attributes.

3. Visualization of the low-level text embeddings for different types of scenes

In this section, we qualitatively shown that the low-level text embeddings for different types of scenes are distinctly separable in the embedding space. We discuss three low-level attributes, i.e., luminance, color, and details, respectively.

For the luminance, we manually categorized the videos from our test subset, which contains 233 videos covering different scenes, into four groups: (1) dark-dominated scenes, (2) bright-dominated scenes, (3) mixed scenes containing both dark and bright regions, and (4) normally illuminated scenes. Specifically, we obtained 40 videos with distinct scenes for the dark-dominated category, 40 for the bright-dominated category, 10 for the mixed category, and 30 for the normally illuminated category, resulting in a total of 120 videos. The remaining 113 videos were excluded due to the difficulty of manual categorization. Representative examples of these videos are shown in Fig. 7, and Fig. 8 further presents the t-SNE visualization of the corresponding luminance text embeddings based on this categorization. As shown in Fig. 8, the embeddings corresponding to dark-dominated, bright-dominated, dark- and bright-dominated, and normally illuminated scenes are clearly well separated in the embedding space. These results demonstrate that the generated luminance text embeddings for different scene types are distinctly separable, thereby demonstrating the effectiveness of

our model in capturing luminance-related characteristics.

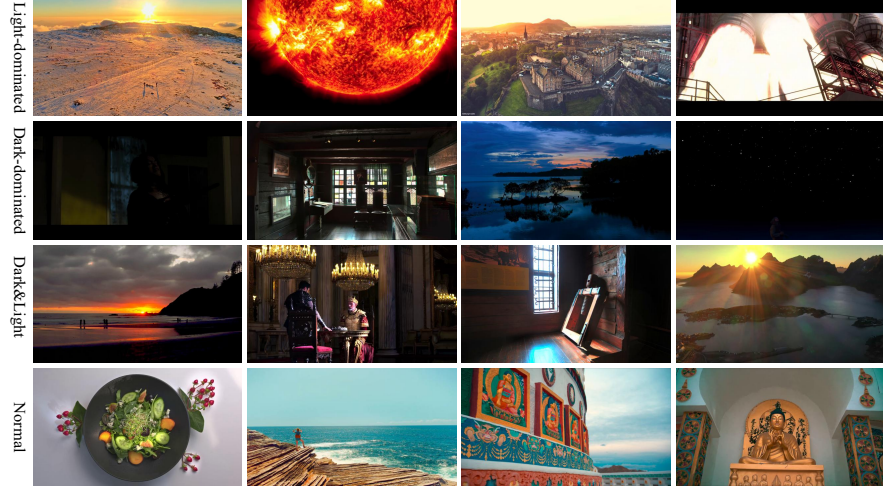


Figure 7: Video examples of scene classification based the luminance.

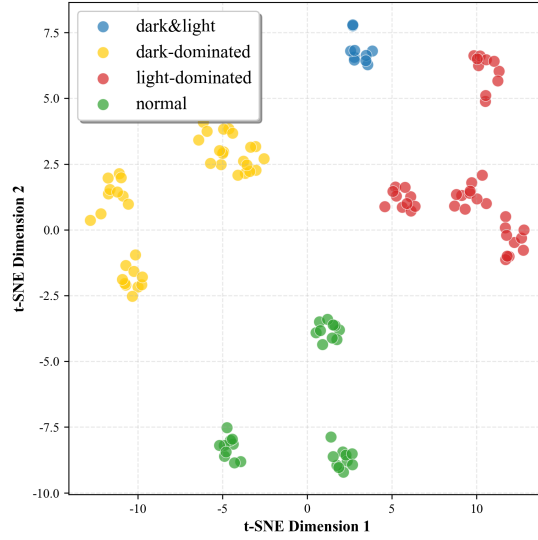


Figure 8: T-SNE figure of luminance text embeddings for different types of scenes.

For the color, we manually categorized the videos from our test subset into five groups: (1) blue-dominated scenes, (2) green-dominated scenes, (3) red-dominated scenes, (4) yellow-dominated scenes, and (5) mixed-color scenes (dominated by two or more colors). Specifically, we obtained 10 videos with

distinct scenes for the blue-dominated category, 20 for the green-dominated category, 10 for the red-dominated category, 20 for the yellow-dominated category, and 60 for the mixed-color category, resulting in a total of 120 videos. The remaining 113 videos were excluded due to the difficulty of manual categorization. Representative examples are shown in Fig. 9, and Fig. 10 presents the t-SNE visualization of the corresponding color text embeddings based on this categorization. As illustrated in Fig. 10, videos dominated by a single color are clearly separated from mixed-color videos. Moreover, perceptually similar colors tend to cluster closer together; for example, blue-dominated and green-dominated scenes lie nearer to each other in the embedding space. These results indicate that the generated color text embeddings for different scene types are distinctly separable, demonstrating the effectiveness of our model in capturing color-related characteristics.

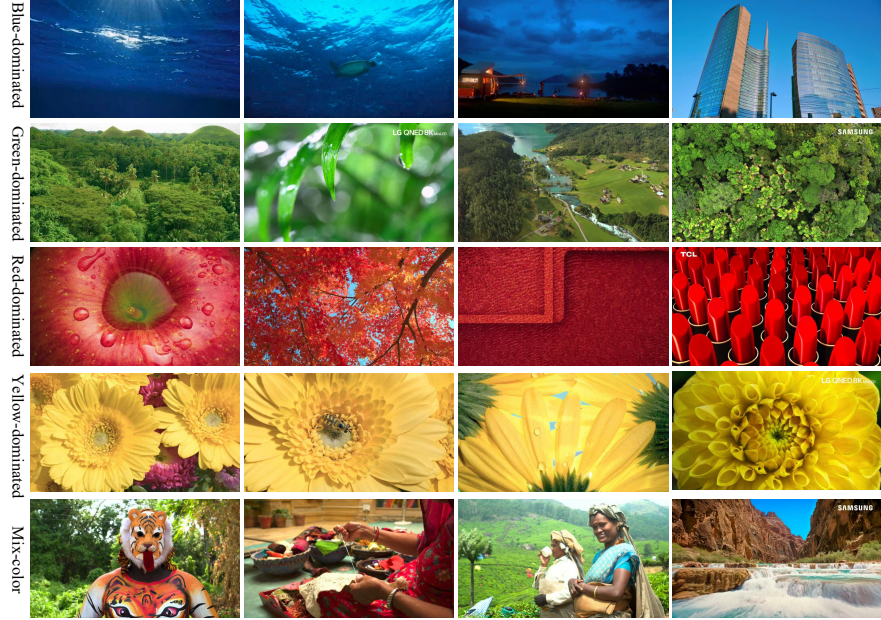


Figure 9: Video examples of scene classification based the color.

For the details, we manually divide the videos from our test subset into two categories: (1) smooth-region-dominated scenes and (2) detail-rich scenes. Specifically, we obtained 30 videos with distinct scenes for the smooth-region-dominated category and 50 for the detail-rich category, resulting in a total of 80 videos. The remaining 153 videos were excluded due to the difficulty of manual categorization. Representative examples of these videos are shown in Fig. 11. Fig. 12 further presents the t-SNE visualization of the corresponding detail text embeddings based on this categorization. As shown in Fig. 12, the embeddings generated from smooth-region-dominated videos are well separated from those

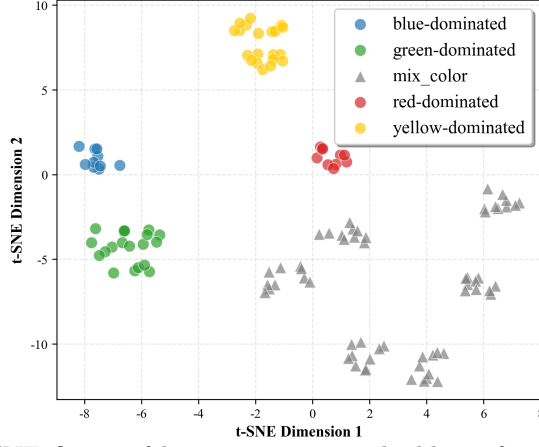


Figure 10: T-SNE figure of luminance text embeddings for different types of scenes.

of detail-rich videos in the embedding space.



Figure 11: Video examples of scene classification based the color.

To intuitively demonstrate the variations in generated captions across diverse video content and compression levels, we present an illustrative examples in Fig. 13. In Fig. 13, we highlight the most prominent differences in red font. As observed from Fig. 13, the captions generated from distinct video content and compression levels exhibit substantial discrepancies. For instance, compression captions yield completely distinct descriptions when the corresponding compression type and level vary.

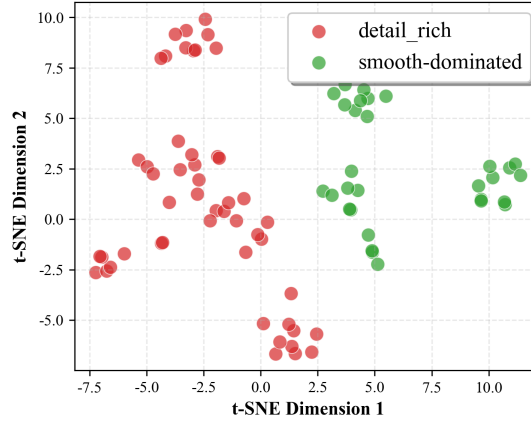


Figure 12: T-SNE figure of luminance text embeddings for different types of scenes.

	VS	
Compression Caption: The video is compressed using H.264 as the compression encoding standard, with the quality control parameter set to 47. H.264 is recognized for its high compression efficiency and is commonly utilized in online video streaming and Blu-ray discs. In H.264, quality can be controlled using either the Constant Rate Factor (CRF) or the Quantization Parameter (QP), allowing for flexibility in balancing file size and visual fidelity. By choosing a QP of 47, the video loses most of its details, with significant compression artifacts, greatly diminishing the visual experience. But this setting ensures transmission with minimal file size.	VS	Compression Caption: The video is compressed using VP9 as the compression encoding standard, with the quality control parameter set to 32. VP9 is celebrated for its open-source nature and high compression efficiency, making it widely applied in streaming services like YouTube for high-resolution video playback. In VP9, quality control is primarily managed through the use of the Constant Rate Factor (CRF). At a CRF of 32, the video retains high quality, with clear details and natural colors, making it a good balance between file size and visual quality. However, while the file size is minimal and transmission is fast, the visual quality suffers greatly.
Semantic Caption: The image showcases a breathtaking view of a mountainous landscape, with a large mountain range visible in the background. The scene is captured from a high vantage point, possibly from a plane or a mountain top, providing an aerial perspective of the landscape. The mountain range is characterized by its steep and rugged terrain, with peaks and valleys visible throughout the scene. The sky is filled with clouds, adding to the dramatic and captivating nature of the image. The overall visual narrative and storytelling effect of the image is a sense of awe and admiration for the beauty and grandeur of the natural world.	VS	Semantic Caption: The image depicts a night scene in a city with a large red building as the main focus. The building is illuminated with red lights, creating a vibrant atmosphere. There are several traffic lights in the scene, with one located near the center and another towards the right side of the image. In addition to the main building, there are two other buildings visible in the background, one on the left side and another on the right side. The presence of multiple traffic lights suggests that this area is a busy intersection or a commercial district. The overall visual narrative and storytelling effect of the image convey a sense of bustling urban life and nighttime activity.
Luminance Caption: The overall brightness of this image is high, with a relatively bright color tone. The brightness of the mountains is high, and the details of the mountains are clear. The brightness of the sky is also high, and the blue sky and white clouds are clearly visible. The overall brightness of the image is high, and the details of the mountains are clear. Therefore, the quality of this image is good.	VS	Luminance Caption: The overall brightness of this image is relatively low, with a slight overexposure in the bright areas. The main subject, the building, is underexposed, resulting in a loss of texture details. The composition is average, and the colors are not vibrant. Therefore, the quality of this image is poor.
Color Caption: The image has accurate exposure with no obvious overexposed or underexposed areas. The focus is accurate, and the picture is clear with no noticeable noise. The colors are rich, and the lighting is sufficient. The texture details are clear, and the composition is centered. The main subject, which is the central mountain, is clearly defined. Therefore, the quality is good.	VS	Color Caption: The overall tone of this image is warm, with a strong red color. The red color is not overly saturated, but it is very vibrant. The image is well-lit, with clear outlines of the buildings and the red lanterns. The lighting is soft, and the colors are harmonious. The composition is good, with the main subject being highlighted. The image has a strong artistic sense, and the colors are well-coordinated. Therefore, the quality of this image is good.
Detail Caption: The texture details of the mountains in the image are rich and clear, with distinct layers and a clear sense of direction. The contrast between the mountains and the sky is strong, and the colors are vibrant. The overall composition is visually appealing, indicating that the quality of this image is good.	VS	Detail Caption: The texture details on the surface of the building in this image are rich and clear, with a clear sense of directionality. The lighting is well-balanced, and the overall composition is visually appealing. Therefore, the quality of this image is good.

Figure 13: Visualization of Generated Captions. In the figure, the left video example is from video #16 (H.264 compression, QP=42) in our test subset, the right video example is from video #24 (VP9 compression, QP=37) in our test subset.

In summary, the low-level text embeddings generated by our model for different types of scenes are distinctly separable in the embedding space, demonstrating the effectiveness of our model in capturing low-level visual characteristics.

C More details of CMA and ITM network

1. More details of Branch-S

In the Fig. 14, we additionally provide compression captions corresponding to three other compression types, namely H.264, H.265, and VP.

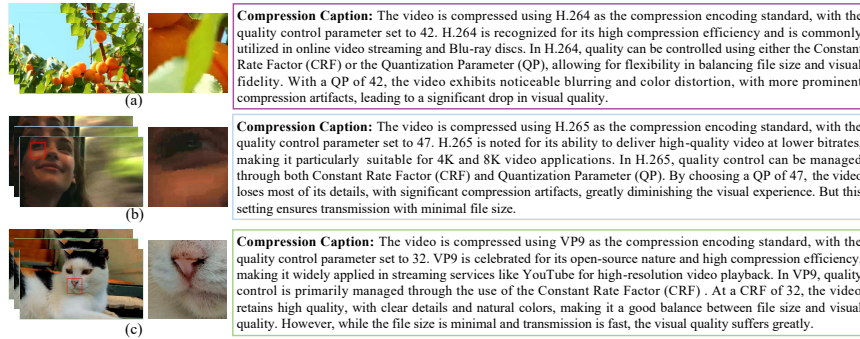


Figure 14: Examples of compression captions produced by the CMA. (a) Video compressed by H.264 with the quality control parameter 42. (b) Video compressed by H.265 with the quality control parameter 47. (c) Video compressed by VP9 with the quality control parameter 32.

2. More details of ITM-network

Fig. 15 and 16 present the LA-MLP and BIAM modules of the ITM network, respectively. Specifically, the BIAM is constructed using a sequence of convolutional (Conv), fully connected (FC), and LSTM layers, while the LA-MLP comprises several linear layers with a softmax activation function.

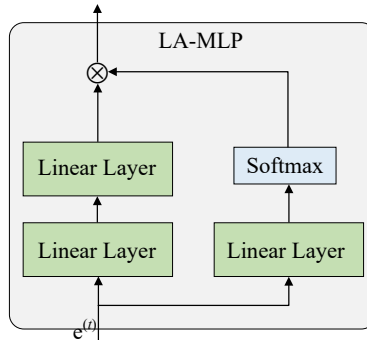


Figure 15: LA-MLP framework.

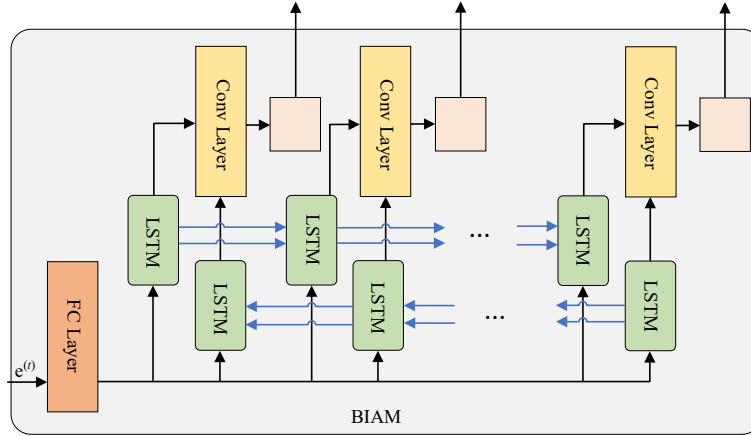


Figure 16: BIAM framework.

3. More details of SDR video descriptor

In Table 1, we present a comparison among the three branches. Specifically, “Tool&MLLM” corresponds to the process in Branch-C, where a tool (e.g., Elecard StreamEye) is first used to extract the compression information of the video, and an MLLM is subsequently employed to generate explanatory descriptions of this compression information. The output dimension of Branch-L is , as it corresponds to three types of low-level attributes, namely luminance, color, and details.

Table 1: Comparison of different branches

	Information Sources	Description Generation Methods	Output dimensions
Branch-C	video metadata	Tool&MLLM	512
Branch-S	video	MLLM	512
Branch-L	video	MLLM	512×3

D More details of database

1. Several representative scene examples from our database

In this section, we present several representative scene examples from our database, as shown in Fig. 17.

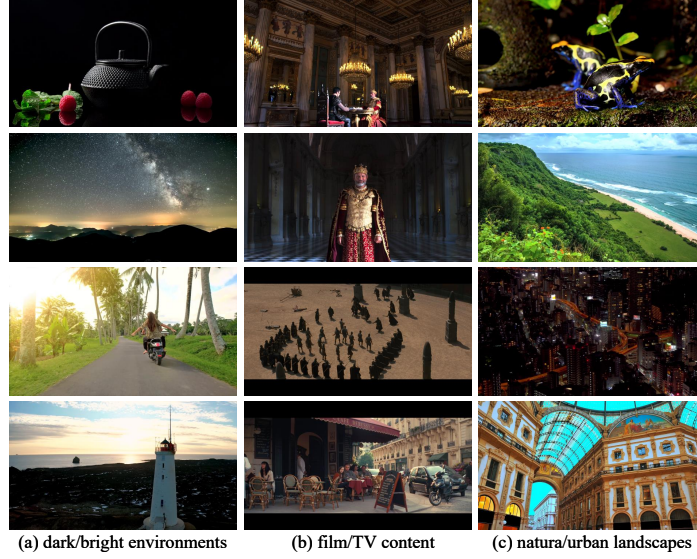


Figure 17: Several representative scene examples from our database.

2. Data diversity analysis

To quantitatively analyze the data diversity, we randomly sampled a frame from each HDR clip and measured the frame by calculating its average luminance level (ALL) (7), average saturation level (ASL) (7), color fullness (CF) (8), and spatial information (SI) (9). Similarly, we also calculated these four

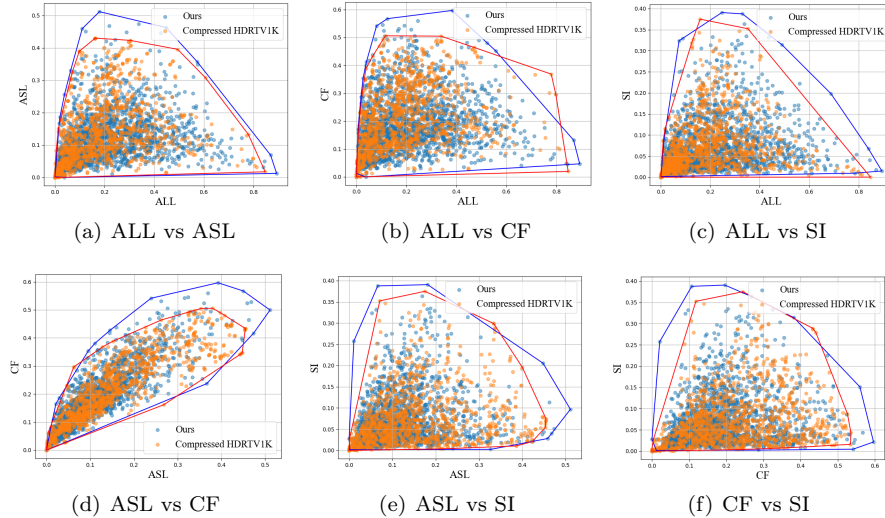


Figure 18: Comparison of our database with the C-HDRTV1K dataset based on six convex hull diagrams. In the figure, the HDR frames from our database are marked by blue dots while the ones from C-HDRTV1K are shown in orange.

measurements on the HDR frames of the C-HDRTV1K dataset. In Fig. 18, convex hull diagrams of the frames from our database and C-HDRTV1K are plotted based on two out of four measurements. From the figure, we can see that our database is with a broader coverage in luminance, color, and spatial information than the C-HDRTV1K dataset.

However, despite its diversity, our database is currently limited to four conventional compression standards. It does not encompass more recent codecs such as VVC, AV1, or neural codecs, nor does it cover more complex compression scenarios, such as variable bitrate (VBR). In future work, we plan to construct a more comprehensive dataset that includes a broader range of compression types to further enhance the generalization of our database.

E Implementation for our model

Some implementation details that are not mentioned in Section 4 of our manuscript will be provided in the following. In the part of SDR video descriptor, the maximum number of tokens for MLLM-generated text is set to 248 to fit the input length constraint of the LongClip. The Branch-C requires a whole video clip to encode the compression information, while the remain two branches works in a frame-wise manner. In the part of global mapping, the feature channel C is set to 128, the number of DRBs is set to 4, and the momentum parameter α in Eq. (2) is set to 0.9. In the part of local compensation, the U-Net architecture consists of three 3×3 convolutional layers and three 3×3 deconvolutional layers. In these layers, the stride of convolution operation is set to 2. In the loss function, the balance parameter between the inter-frame loss and the frame-wise loss is set to 0.05 empirically.

The prepared data presented in Section 3.4 is employed to train and test the proposed model. We initially downloaded 120 paired SDR-HDR videos from YouTube. Then, following the above metadata-based and visual quality-control strategy, we manually selected 70 high-quality SDR-HDR video pairs for constructing our dataset. Specifically, for each HDR video collected from YouTube, we first checked its metadata and retained only videos that are in 10-bit format, use the BT.2020 color space and the PQ transfer function, and have a bitrate higher than 20,000 kbps. We then displayed each candidate video on an ASUS PA32UCX HDR monitor and invited five volunteers with experience in compressed video and HDR content quality assessment to evaluate its visual quality under the same viewing conditions. A video was included in our dataset only when all five volunteers confirmed that it contained no visually noticeable compression artifacts, severe banding, abnormal tone mapping, highlight clipping, or color distortion. Through this screening process, we excluded low-quality HDR sources and ensured that the selected videos have valid HDR format and high visual quality. The 70 selected videos contain frequent scene content changes. Therefore, we manually divided these videos into 2,330 video clips, ensuring that each clip contains no scene change and no duplicate content. We then split these video clips into training, validation, and testing subsets with

a ratio of 8:1:1. After this split, the SDR clips in the training, validation, and testing subsets were recompressed separately to generate the compressed SDR videos. After recompression, the dataset contains 298,240 training pairs, 37,280 validation pairs, and 37,280 testing pairs. To determine the tunable parameters in the proposed model, we adopt a three-stage training procedure, as mentioned in Section 3.5. In all the stages, the batch size is set to 4. To make the training efficient, each 4K frame is cropped into 32 patches with resolution of 480×480 pixels. The cropped patches instead of the original frames serve as the inputs of global mapping and local compensation in the training. Notably, the input of the SDR descriptor part is kept uncropped to ensure the validity of the compression, semantic, and low-level information in SDR videos. As the SDR descriptor part contains no tunable parameter, we can obtain the descriptive feature $e^{(t)}$ of each training patch prior to the training. And to improve training efficiency, the descriptive feature $e^{(t)}$ are generated in advance. Besides, the model is initialized by the Xavier method (10) and optimized by the Adam method (11) ($\beta_1 = 0.9$ and $\beta_2 = 0.99$). In the first and second stages of training, the learning rate is set to 2×10^{-4} , and the learning rate is halved for every 30 epochs. In the third stage, we set the learning rate to 5×10^{-5} , which is halved for every 20 epochs. Notably, the validation occurs every 5 epochs during training, with all parts maintaining 4K resolution inputs during validation. In each stage, the validation is conducted every 5 epochs and the training stops when the ITM performance on the validation subset stabilizes. When validating and testing the ITM model, all the inputs with 4K resolution are kept uncropped.

F Supplementary of experiments

1. Flicking scores and qualitative comparisons of temporal consistency

We conducted a subjective quality assessment by involving 20 participants to grade the annoyance degree of HDR videos in terms of flicking on five levels: “Very Annoying”, “Annoying”, “Slightly Annoying”, “Perceptible but not Annoying”, and “Imperceptible”. The experimental environment and data processing are kept the same as (12). The mean opinion scores of this study, named as Flicking Scores, are listed in the last row of Table 2, which also demonstrate the effectiveness of our model. For qualitative comparison, to demonstrate the visual stability, we provide an example in Fig. 19, where we show the pixel values at a fixed spatial position along with time. As shown in Fig. 19, the pixel value transitions produced by our model are significantly smoother compared to those of the DFT. This observation quantifies the superior temporal consistency of our model.

Table 2: Quantitative Comparisons between Our Model and SOTA Ones

Model	KPN-MFI	ICTCPNet	DFT	KPNet	CAVSR	CAVSR+ITM	LLMRA	LLMRA+ITM	Ours
Flicking Score $\uparrow$	<u>4.27</u>	3.92	2.91	3.65	2.84	3.61	3.57	3.24	4.41

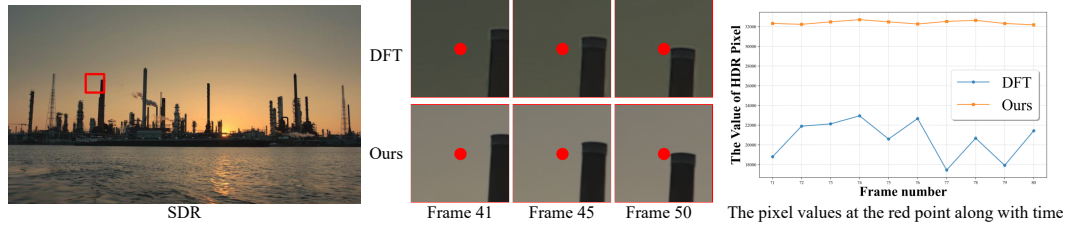


Figure 19: Qualitative Comparison between Ours with DFT.

2. Computational costs of different parts

To provide a clearer understanding of the computational costs, we include a comprehensive breakdown of the inference time for each stage in Table 3. As shown in Table 3, the MLLM inference (i.e., caption generation) is indeed the primary bottleneck, accounting for 11.0680s per 4K frame. In contrast, the remaining components of our ITM pipeline, including text encoding, global mapping, and local compensation, are highly efficient, requiring only 1.0746s in total. However, ITM field encompasses two primary application scenarios: 1) converting existing SDR content to HDR ones; and 2) Real-time terminal display, which requires instantaneous SDR-to-HDR conversion. Notably, while the latter imposes stringent constraints on inference latency, the former allows for a more flexible time budget. To the best of our knowledge, only look-up table (LUT)-based ITM methods can achieve real-time terminal display, while most existing deep learning-based ITM models struggle to meet the stringent temporal requirements of real-time applications. Consequently, our work primarily targets the first application scenario.

Table 3: Computational cost comparison of different parts in our model

	Caption generation	Text encoding	Global mapping	Local compensation
Time(s)	11.0680	0.2312	0.1707	0.6727
FLOPs	3428.24G	24.96G	477.45G	917.08G
Complexity	$O(N^2)$	$O(L^2)$	$O(N)$	$O(N^2)$

To explore lightweight MLLM variants, we compare several popular lightweight MLLMs, as shown in Table 4. Notably, the values reported in the table, including runtime, GPU memory consumption, and FLOPs, are measured for the caption generation part. To convenience, we directly replace the MLLM in our model with these variants without performing task-specific fine-tuning on the Q-instruct dataset (4). As a result, their ability to capture low-level attribute information is limited, leading to lower PSNR compared to our model. Among the evaluated MLLM variants, Deepseek-VL-1.3B achieves a favorable trade-off among runtime, GPU memory consumption, and FLOPs. This suggests that replacing the MLLM in our model with this model can substantially reduce the overall computational cost and significantly improve the practical deployability of our model. It is worth noting that we only considers a subset of efficient

MLLMs; other lightweight and high-performing MLLMs also could be explored in future work.

Table 4: Computational Cost Comparison of Different MLLM Variants

MLLM variants	Runtime/(s)	GPU memory/(G)	FLOPs/(G)	PSNR/(dB)
MobileVLM-V2-1.7B (13)	8.9920	4	2410.48	36.47
Qwen2-VL-2B (14)	8.2216	5.2	3047.82	36.62
Deepseek-VL-1.3B (15)	7.2796	5.1	2458.59	36.54
Mplug-Owl2-llama2-7B(Ours)	11.0680	16.3	3428.24	37.28

We frankly acknowledge that the significant latency of MLLM inference limits the current practicality of our pipeline for real-time applications. Therefore, our future work will focus on three directions, as suggested by the reviewer, to enable efficient video processing. First, we will investigate the adoption of lightweight MLLM variants (e.g., Deepseek-VL-1.3B) to substantially reduce inference latency. Second, we will explore a temporal caption caching mechanism. Since adjacent video frames often exhibit strong temporal consistency, captions can be generated only for key frames (e.g., at scene cuts or fixed intervals) and reused for neighboring frames, thereby significantly reducing redundant computation. Third, we will study knowledge distillation for model acceleration, in which the captioning knowledge of large MLLMs is transferred to lightweight networks, enabling efficient caption prediction without relying on heavy models during inference. In addition, for offline or non-real-time applications, captions can be pre-generated as a preprocessing step to further alleviate runtime overhead. We believe that integrating these strategies will significantly improve the practical efficiency and real-world applicability of our model.

3. Result of different QP values and compression types

We present quantitative results under two configurations: (1) different QP values under H.264 compression, and (2) different compression types at QP=42. Due to space limitations, the details of the two configuration results are provided in Supplementary B. Overall, our model achieves the best results in all cases. It demonstrates that our model can reliably reconstruct high-quality HDR outputs under various compression types and QPs. as shown in Table 5 and Table 6, respectively.

Table 5: Comparison of Different QP Values for H.265 Compression

Comp.types	Model	PSNR $\uparrow$	SSIM $\uparrow$	HDR-VDP3 $\uparrow$	ΔE_{ITP} $\downarrow$
32	KPNet	38.39	0.9636	8.210	8.36
	ICTCPNet	38.20	0.9628	8.169	8.88
	LLMRA+ITM	<u>38.69</u>	<u>0.9655</u>	<u>8.237</u>	<u>8.09</u>
	Ours	39.11	0.9672	8.278	7.92
37	KPNet	36.93	0.9532	7.822	9.75
	ICTCPNet	36.77	0.9516	7.773	10.05
	LLMRA+ITM	<u>37.14</u>	<u>0.9547</u>	<u>7.813</u>	<u>9.40</u>
	Ours	37.68	0.9553	7.842	9.22
42	KPNet	35.10	0.9343	7.320	11.10
	ICTCPNet	34.96	0.9338	7.272	11.79
	LLMRA+ITM	<u>35.28</u>	<u>0.9368</u>	7.277	11.18
	Ours	35.56	0.9385	7.323	10.85
47	KPNet	32.83	0.9083	6.694	14.25
	ICTCPNet	32.70	0.9062	6.634	14.72
	LLMRA+ITM	<u>33.01</u>	<u>0.9105</u>	6.612	<u>13.89</u>
	Ours	33.45	0.9113	<u>6.658</u>	13.67

Table 6: Comparison of Different Compression Types under QP=42

Comp.types	Model	PSNR		SSIM		HDR-VDP3		ΔE_{ITP}	
		mean	std	mean	std	mean	std	mean	std
AVS2	KPNet	35.46	3.8746	0.9440	0.0574	7.469	2.8437	10.81	2.4210
	ICTCPNet	35.27	3.7521	0.9445	0.0521	7.485	2.8619	11.71	2.5814
	LLMRA+ITM	35.75	3.4417	0.9479	0.0490	7.517	2.7594	<u>10.49</u>	2.3480
	Ours	36.03	3.1448	0.9502	0.0479	7.562	2.7241	10.05	2.0133
H.264	KPNet	34.72	4.0141	0.9268	0.0774	7.156	2.9774	<u>11.60</u>	2.5547
	ICTCPNet	34.67	3.6548	0.9280	0.0697	7.160	3.1527	12.19	<u>2.4731</u>
	LLMRA+ITM	<u>35.07</u>	<u>3.4773</u>	<u>0.9304</u>	<u>0.0614</u>	<u>7.165</u>	<u>2.8104</u>	11.64	2.5202
	Ours	35.14	3.2809	0.9321	0.0570	7.215	2.6521	11.48	2.0118
H.265	KPNet	35.10	3.8752	0.9343	0.0637	<u>7.320</u>	3.0947	<u>11.10</u>	2.6855
	ICTCPNet	34.96	3.9217	0.9338	<u>0.0594</u>	7.272	3.3860	11.79	<u>2.4820</u>
	LLMRA+ITM	<u>35.28</u>	<u>3.7740</u>	<u>0.9368</u>	0.0626	7.277	<u>2.9110</u>	11.18	2.5022
	Ours	35.56	3.4311	0.9385	0.0474	7.323	2.8290	10.85	2.0167
VP9	KPNet	38.79	3.5430	0.9660	0.0425	8.312	2.5487	8.09	2.2173
	ICTCPNet	38.64	3.3613	0.9662	<u>0.0301</u>	8.258	2.6302	8.59	2.1590
	LLMRA+ITM	<u>39.09</u>	<u>3.0271</u>	<u>0.9685</u>	0.0430	<u>8.331</u>	<u>2.4733</u>	<u>7.87</u>	<u>2.0376</u>
	Ours	39.49	2.6606	0.9702	0.0295	8.389	2.3426	7.51	2.0091

Table 7: Average PSNR and SSIM values of each codec at QP=42

Compr.types	AVS2	H.264	H.265	VP9
PSNR $\uparrow$	33.2957	33.7680	33.6635	41.0512
SSIM $\uparrow$	0.8912	0.8447	0.8922	0.9446

Here we only list the results from some representative models in Table 5. As shown in Table 5, the four evaluation metrics of all models decrease as the QP value increases. This clearly demonstrates that the more severe the compression artifacts, the greater their negative impact on the ITM task. In Table 6, “mean” denotes the average value, and “std” represents the standard deviation. As shown in Table 6, all models perform relatively poorly when the compression type is H.264, H.265, or AVS2. This may be because, for these codecs, QP=42 corresponds to a relatively severe level of compression, which significantly degrades ITM performance. In contrast, when the compression type is VP9 at QP=42, all models achieve comparatively better results. By examining most of the SDR videos generated with different compression types at QP=42, we observed that the SDR videos compressed using VP9 suffer less from compression artifacts. In addition, we present the average PSNR and SSIM values of the compressed SDR frames against the original SDR frames for each codec at QP=42 in Table 7. As demonstrated in Table 7, the SDR frames compressed with VP9 achieve SOTA performance, including an average PSNR of 41.05 dB and an average SSIM of 0.9446, which corroborates our explanation that VP9 suffers less from compression artifacts, making the ITM task easier. This likely explains why all models achieve better ITM results on VP9 compressed videos compared to videos compressed with other types. Furthermore, our model attains SOTA results with respect to standard deviation (i.e., std), demonstrating that it provides more consistent and stable performance than competing models. Overall, our model achieves the best results in all cases. It demonstrates that our model can reliably reconstruct high-quality HDR outputs under various compression types and QPs.

4. Quantitative comparisons in public dataset

To demonstrate the robustness of our model, we extend our experiments into public dataset, i.e., C-HDRTV1K dataset. We retrained and evaluated our model alongside several state-of-the-art models on this dataset. As shown in Table 8, our model consistently achieves SOTA performance across all metrics, including PSNR, SSIM, HDR-VDP3, and ΔE_{ITP} . The SOTA results on this public dataset demonstrate the robustness and generalization capability of our model.

Table 8: Quantitative Comparisons in C-HDRTV1K

Model	KPN-MFI	ICTCPNet	DFT	KPNet	CAVSR	CAVSR+ITM	LLMRA	LLMRA+ITM	Ours
PSNR $\uparrow$	32.13	32.51	32.08	32.37	32.01	32.31	32.81	<u>33.06</u>	33.21
SSIM $\uparrow$	0.8918	0.8926	0.8891	0.8923	0.8913	0.8925	0.8933	<u>0.8945</u>	0.8953
HDR-VDP3 $\uparrow$	<u>6.745</u>	6.649	6.624	6.681	6.739	6.591	6.735	6.736	6.814
$\Delta E_{ITP} \downarrow$	16.32	15.11	16.24	15.84	16.71	15.73	14.82	14.24	13.87

5. Color consistency comparison

Since the subjective results, e.g. Fig. 8 in the manuscript, cannot clearly demonstrate the superiority of our model in terms of color consistency, we further analyze this aspect using color histograms. To improve clarity, Fig. 20

presents only a few representative examples. As illustrated in Fig. 20, the color histogram and gamut distribution produced by our model more closely match those of GT, demonstrating its superior ability to preserve color consistency.

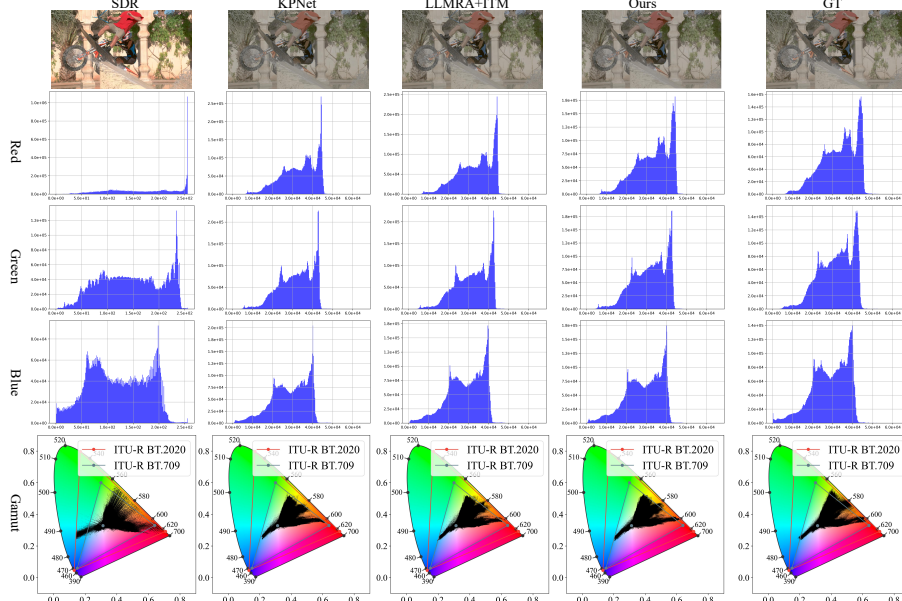


Figure 20: Color consistency comparison.

6. Additional ablation study

6.1 Qualitative comparisons of Section 4.3.2 and 4.3.3 in our manuscript

Fig. 21 and 22 present the qualitative comparisons for Sections 4.3.2 (Impact of Video Information Description) and 4.3.3 (Impact of Different Embedding Strategies) of our manuscript, respectively. As shown in Fig. 21, the “N/A” variant exhibits noticeable compression artifacts. In contrast, both “Compr. inf” and “Ours” effectively suppress such artifacts, with “Ours” producing results that are closest to the GT. These observations demonstrate the effectiveness of incorporating video information descriptions into the ITM network.

As shown in Fig. 22, all variants except “Ours” exhibit noticeable color shifts, and the “w/o–w/o” further produces evident artifacts. These results demonstrate the effectiveness of our embedding strategy.



Figure 22: Qualitative comparisons of different embedding strategies.

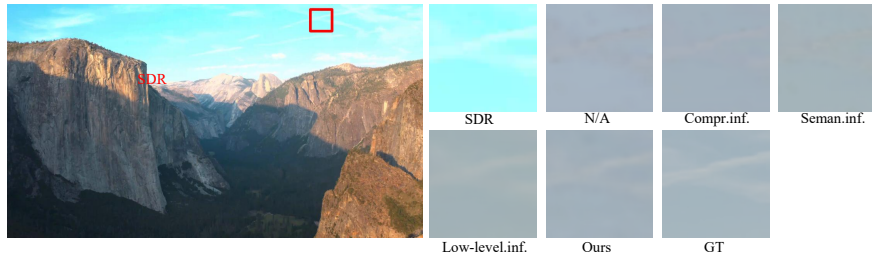


Figure 21: Qualitative comparisons of different text descriptions.

6.2 Impact of hyperparameter

To determine more suitable hyperparameter settings, such as momentum parameter α and the loss balance weight λ , we performed the following ablation studies. Specifically, for experimental efficiency, PSNR is utilized as the core metric for empirical parameter sensitivity analysis, and the corresponding results are tabulated in Table 9 and 10. These tables demonstrate that our model attains SOTA performance with $\lambda=0.05$ and $\alpha=0.9$. We thus utilize these values as the definitive hyperparameters of our model.

Table 9: Sensitivity of loss balance weight λ

λ	0.01	0.03	0.05
PSNR $\uparrow$	36.82	37.14	37.28

Table 10: Sensitivity of momentum parameter α

α	0.1	0.3	0.5	0.7	0.9
PSNR $\uparrow$	34.23	35.48	36.57	37.02	37.28

6.3 Impact of different instructions

To demonstrate the effectiveness of our instruction design, we further conduct a brief ablation study. Specifically, we replace the instructions for generating low-level attributes with simpler prompts, such as “Describe the texture

detail information of this image”, while keeping the instructions for generating semantic captions unchanged. We refer to this model variant as Ours-ins. A simple example is shown in Fig. 23. As illustrated in Fig. 23, simplified instructions lead to the generation of vague captions. Then, the quantitative results are reported in Table 11, and the qualitative comparisons are presented in Fig. 24. As shown in Table 11, “Ours” achieves SOTA performance. As illustrated in Fig. 24, the “Ours-ins” produces noticeable artifacts, whereas “Ours” exhibits no visually perceptible artifacts. This indicates that using simplified instructions leads the MLLM to generate vague descriptions of low-level attributes, which in turn degrades the model’s ability to faithfully restore such attributes. These results further demonstrate the effectiveness of our instruction design.

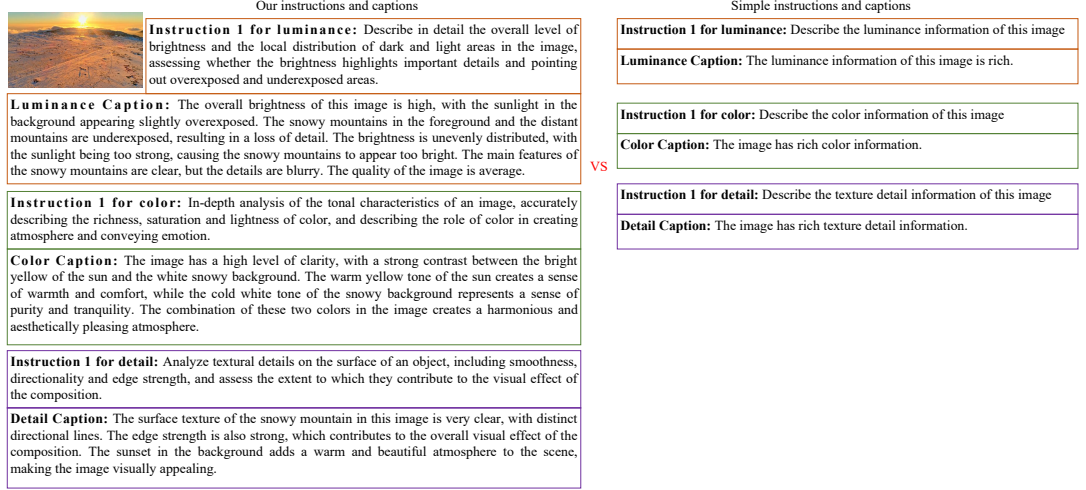


Figure 23: An example of instructions and captions for Ours and Ours-ins.

Table 11: Quantitative Comparisons between Ours with Ours-ins

Model	PSNR $\uparrow$	SSIM $\uparrow$	HDR-VDP3 $\uparrow$	ΔE_{ITP} $\downarrow$
Ours-ins	36.55	0.9487	7.553	9.64
Ours	37.28	0.9532	7.806	9.37

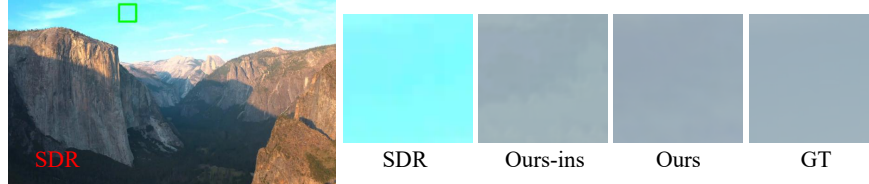


Figure 24: Qualitative Comparisons between Ours with Ours-ins.

We acknowledge that while our current instructions can effectively prompt the MLLM for accurate low-level attribute descriptions, there may exist even more optimized phrasing strategies, which we intend to explore in future work. And we also believe that numerous other low-level attributes beneficial to ITM task remain unexplored, and we will prioritize the investigation of these attributes in our future work.

6.4 Impact of feature injection positions

To demonstrate the effectiveness of our feature injection, we have designed a brief ablation study, the results of which are presented in Table 12 and Fig. 25. In Table 12 and Fig. 25, the "Encoder & Decoder" denotes the configuration where the descriptive feature $e^{(t)}$ is injected into both the encoder and decoder of the adaptive local compensation. As shown in Table 12, our model achieves slightly lower PSNR and SSIM than the "Encoder & Decoder" variant, but performs slightly better in terms of HDR-VDP-3, ΔE_{ITP} , and runtime. As illustrated in Fig. 25, there is no noticeable visual difference between our method and the "Encoder & Decoder" variant. Therefore, we choose to inject the descriptive feature $e^{(t)}$ only into the decoder. Nevertheless, injecting $e^{(t)}$ into both the encoder and decoder remains a feasible alternative.

Table 12: Ablation Results on Feature Injection Positions

Model	PSNR $\uparrow$	SSIM $\uparrow$	HDR-VDP3 $\uparrow$	$\Delta E_{ITP} \downarrow$	Runtime
Encoder & Decoder	37.41	0.9537	7.789	9.45	12.47s
Ours	37.28	0.9532	7.806	9.37	12.14s

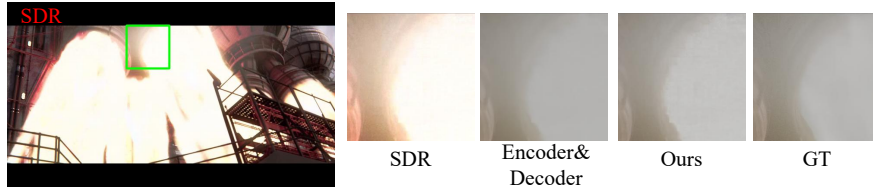


Figure 25: An example of visualization comparisons.

6.5 Impact of the hallucination of MLLM

To demonstrate the impact of MLLM’s hallucination, we conducted a simple ablation study by transforming the generated captions from the test subset into inaccurate or vague descriptions. Specifically, we employed the MLLM to deliberately distort the original captions by providing it with an instruction to generate inaccurate or vague descriptions. Two illustrative examples are presented in Fig. 26 and Fig. 27. Specifically, "Semantic Caption 2" is the inaccurate description in Fig. 26 and "Semantic Caption 3" is the vague description in Fig. 27. We then conducted both qualitative and quantitative evaluations to assess the model’s sensitivity to these textual inaccuracies. As shown in Table 13,

when provided with inaccurate descriptions, our model’s performance degrades significantly: the PSNR drops by approximately 1 dB, and the ΔE_{ITP} increases from 9.37 to 10.65; When provided with vague descriptions, the performance exhibits a slight decline. As shown in Fig. 28, inaccurate descriptions lead to noticeable color shifts in the output, whereas vague descriptions do not produce obvious perceptual differences. These results demonstrate that our model is responsive to the linguistic cues provided by the MLLM. Conversely, they also indicate that vague or inaccurate descriptions can indeed negatively affect performance. To mitigate this, we aim to extend the current model into an interactive system and consider the development of verification strategies, aimed at cross-checking the factual accuracy of MLLM-generated descriptions, to be a critical direction for our future work to ensure the robustness of text-driven ITM model.

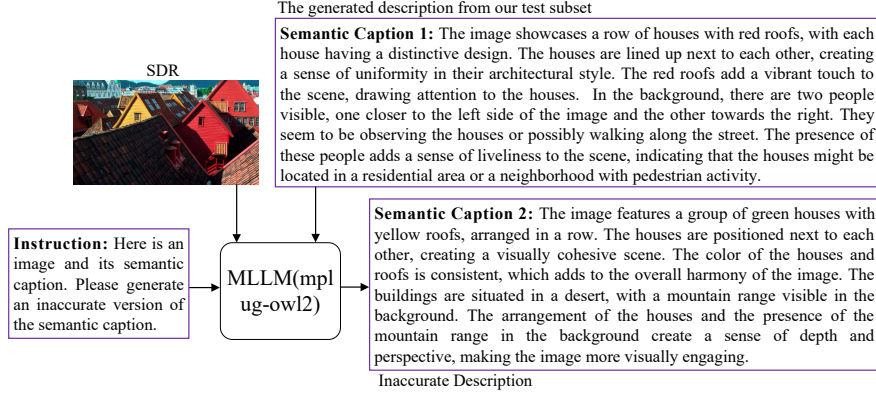


Figure 26: An example of inaccurate description. The same procedure is applied to the remaining captions.

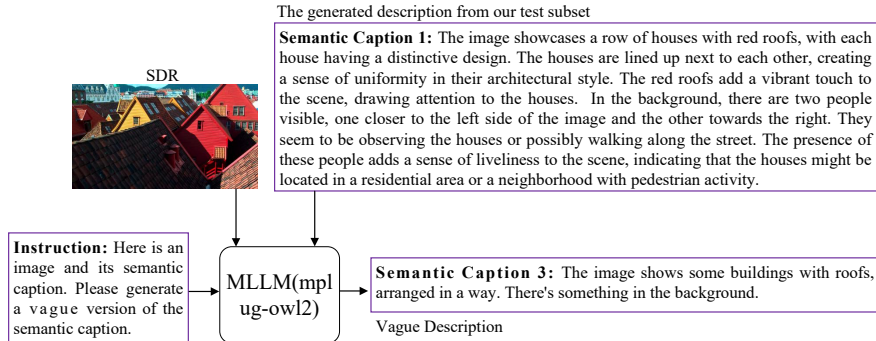


Figure 27: An example of vague description. The same procedure is applied to the remaining captions.

Table 13: Ablation Results on Feature Injection Positions

Descriptions types	PSNR $\uparrow$	SSIM $\uparrow$	HDR-VDP3 $\uparrow$	$\Delta E_{ITP} \downarrow$
Inaccurate description	36.22	0.9511	7.684	10.65
Vague description	36.74	0.9526	7.773	9.89
Ours	37.28	0.9532	7.806	9.37

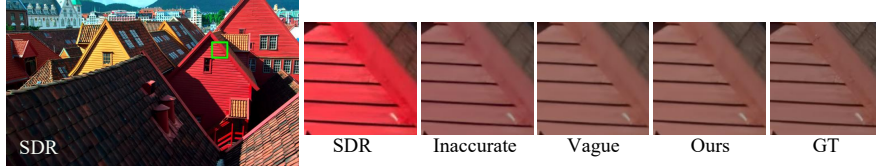


Figure 28: An example of visualization comparisons for different descriptions.

6.6 Impact of data

To evaluate the effect of different databases on the compressed video ITM task, we conduct two experiments: The first is training our model on the C-HDRTV1K database and testing it on our database. The second is training our model on our database and testing it on the C-HDRTV1K database. Table 14 presents the quantitative results for both experiments.

Table 14: Comparison of Different Databases

Train subset	Test subset	PSNR $\uparrow$	SSIM $\uparrow$	HDR-VDP3 $\uparrow$	$\Delta E_{ITP} \downarrow$
C-HDRTV1K	Ours	33.14	0.8950	6.727	14.10
Ours	C-HDRTV1K	37.62	0.9783	7.079	11.18
Ours	Ours	37.28	0.9532	7.806	9.37

As shown in Table 14, the first experiment yields the poorest quantitative results. Meanwhile, the second experiment demonstrates comparable performance to our model’s results, with slightly higher PSNR and SSIM values. This indicates that C-HDRTV1K, containing only one compression type, fails to generalize to our dataset featuring four compression types, underscoring the necessity of constructing our database.

6.7 Impact of the sliding manner

To enhance the temporal consistency of textual descriptions, we adopt a sliding manner to generate both semantic and low-level descriptions. Specifically, this manner enforces a consistent sentence structure for the same type of descriptions. It only updates the corresponding parts of the caption for regions where scene changes occur between the current frame and the previous frame, while keeping the remaining parts unchanged as much as possible. This avoids re-generating descriptions from scratch for every video frame using the MLLM,

thereby preventing large fluctuations in wording and phrasing that would otherwise compromise temporal consistency. In Table 15 and Figure 29, we present the quantitative and qualitative comparisons between the variant without the sliding manner (“w/o sliding manner”), which independently generates descriptions for each video frame using “Instruction 1”, and our model (“Ours”). As shown in Table 15, compared with “Ours”, the “w/o sliding manner” variant exhibits a decrease of approximately 1 dB in tPSNR and an increase of about 0.005 in MABD. The line plots in Figure 29 further demonstrate that “w/o sliding manner” produces significant fluctuations, whereas “Ours” maintains a relatively stable trend. These results demonstrate that generating semantic and low-level descriptions in a sliding manner effectively improves the temporal consistency of our model.

Table 15: Quantitative measures of temporal consistency

	w/o sliding manner	Ours
tPSNR $\uparrow$	37.6971	38.6428
MABD $\downarrow$	0.0154	0.0118

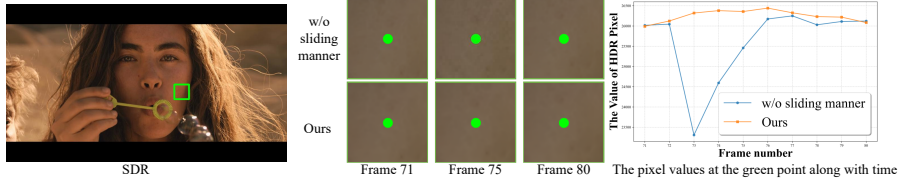


Figure 29: Qualitative Comparison between w/o sliding manner with Ours.

Prior work (3) has shown that the sliding manner can generate high-quality captions. Moreover, as discussed in Section A.1.2, the captions generated on our test subset exhibit almost no hallucinations or erroneous descriptions, suggesting that such issues occur with low probability in our model. In addition, since the sliding manner takes the previous frame, its caption, and the current frame as joint inputs, the MLLM has access to both prior visual information and textual context. This enables the MLLM to gradually correct potential errors from earlier frames during subsequent caption generation, thereby mitigating error accumulation. Notably, since the current caption inherits the previous template and only updates changed regions, prior hallucinations or errors will largely persist uncorrected. While this may reduce overall visual quality, it prevents abrupt textual discontinuities that would lead to more noticeable temporal inconsistencies, which are far more detrimental to perceptual video quality.

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